Jingyun Yang

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Research Interests

I'm enthusiastic about building robot learning frameworks that empower autonomous agents with the capability to perform complex manipulation skills involving deformable and articulated objects, adapt or generalize to unseen scenarios in the real world, and make plans to complete long-horizon tasks.

Education

Stanford University — Stanford, CA
PhD in Computer Science — *September 2022 – Present*

Carnegie Mellon University — Pittsburgh, PA
M.S. in Machine Learning · GPA: 4.11/4.33 — *August 2020 – December 2021*

University of Southern California — Los Angeles, CA
B.S. in Computer Science · GPA: 4.00/4.00 — *August 2016 – May 2020*

Publications

1. **Jingyun Yang***, Congyue Deng*, Jimmy Wu, Rika Antonova, Leonidas Guibas, Jeannette Bohg, "EquivAct: SIM(3)-Equivariant Visuomotor Policies beyond Rigid Object Manipulation", in submission to International Conference on Robotics and Automation (ICRA), 2024.
2. **Jingyun Yang***, Max Sobol Mark*, Brandon Vu, Archit Sharma, Jeannette Bohg, Chelsea Finn, "Robot Fine-Tuning Made Easy: Pre-Training Rewards and Policies for Autonomous Real-World Reinforcement Learning", in submission to International Conference on Robotics and Automation (ICRA), 2024.
3. Rika Antonova*, **Jingyun Yang***, Krishna Murthy Jatavallabhula, Jeannette Bohg, "Rethinking Optimization with Differentiable Simulation from a Global Perspective", Conference on Robot Learning (CoRL), 2022. **(Oral.)**
4. **Jingyun Yang**, Junwu Zhang, Connor Settle, Akshara Rai, Rika Antonova, Jeannette Bohg, "Learning Periodic Tasks from Human Demonstrations", International Conference on Robotics and Automation (ICRA), 2022.
5. Rika Antonova, **Jingyun Yang**, Priya Sundaresan, Dieter Fox, Fabio Ramos, Jeannette Bohg, "A Bayesian Treatment of Real-to-Sim for Deformable Object Manipulation", IEEE Robotics and Automation Letters, 2021.
6. **Jingyun Yang***, Hsiao-Yu Fish Tung*, Yunchu Zhang*, Gaurav Pathak, Ashwini Pokle, Christopher G. Atkeson, Katerina Fragkiadaki, "Visually-Grounded Library of Behaviors for Manipulating Diverse Objects across Diverse Configurations and Views", Conference on Robot Learning (CoRL), 2021.
7. Youngwoon Lee, **Jingyun Yang**, Joseph J. Lim, "Learning to Coordinate Manipulation Skills via Skill Behavior Diversification", International Conference on Learning Representations (ICLR), 2020.
8. Karl Pertsch, Oleh Rybkin, **Jingyun Yang**, Konstantinos Derpanis, Kostas Daniilidis, Joseph J. Lim, Andrew Jaegle, "Keyframing the Future: Keyframe Discovery for Visual Prediction and Planning", Learning for Dynamics and Control (L4DC), 2020.
9. Kien Nguyen, **Jingyun Yang**, Yijun Lin, Jianfa Lin, Yao-Yi Chiang, Cyrus Shahabi, "Los Angeles Metro Bus Data Analysis Using GPS Trajectory and Schedule Data (Demo Paper)", ACM SIGSPATIAL, 2018.

Experience

Stanford Interactive Perception and Robot Learning (IPRL) Lab — Palo Alto, CA
Research Assistant · Advisor: Jeannette Bohg — *Jun. 2021 – Sep. 2021; Feb. 2022 – Present*

- **Equivariant Visuomotor Policies for Generalizable Robot Learning:** Integrated equivariant neural network architectures with imitation learning frameworks and built a pipeline for generalizing visuomotor

policies to unseen scenarios with dramatically different translations, scales, and rotations from those seen in demonstrations. Paper in submission to ICRA 2024.

- **Autonomous Fine-Tuning of Robot Manipulation Policies:** Built a robot learning system that utilizes pre-trained offline RL policies and rewards to achieve autonomous fine-tuning with minimal human supervision. Paper in submission to ICRA 2024.
- **Differentiable Simulators for Efficient Robot Learning:** Experimented with 15 differentiable simulation environments based on 4 different simulation frameworks; examined the quality of gradients in these environments; tested global search and reinforcement learning algorithms on learning policies in these environments. Paper accepted as an oral presentation at CoRL 2022.
- **Active Learning Algorithms for Periodic Robot Manipulation Tasks:** Explored appropriate policy structures for periodic tasks and designed an algorithm based on Bayesian Optimization to learn an optimal policy efficiently. Paper accepted at ICRA 2022.

CMU Professor Katerina Fragkiadaki's Lab Pittsburgh, PA
Research Assistant · Advisors: Katerina Fragkiadaki & Christopher G. Atkeson *Jun. 2020 – Dec. 2021*

- **Efficient Exploration with Kinematics Prediction:** Investigated exploration methods that predict and fine-tune the kinematic structure of a scene while interacting with it using the predicted scene structure.
- **Visual Representations for Generalizable Manipulation:** Designed and implemented a library-based framework that learns visual representations to help robotic agents generalize to varying objects, initial poses, and camera viewpoints. Paper accepted at CoRL 2021.

USC Cognitive Learning for Vision and Robotics (CLVR) Lab Los Angeles, CA
Research Assistant · Advisor: Joseph J. Lim *Aug. 2018 – May 2020*

- **Compositional Imitation Learning:** Researched latent space models that learn useful robot skills from a large set of demonstrations to achieve efficient hierarchical learning of a downstream task.
- **Learning Coordinated Tasks from Rewards:** Designed, implemented, and tested a hierarchical reinforcement learning model that tackles complex tasks of multiple agents by learning composable primitive skills and coordination of the skills. Paper accepted at ICLR 2020.
- **Video Prediction with Temporal Hierarchy:** Developed experiments and designed a downstream planning algorithm for a model that learns to discover informative frames in a future video sequence and represent the video via its keyframes. Paper accepted at L4DC 2020.

Bloomberg L.P. New York, NY
Software Engineering Intern · Supervisor: Luke Shen *May 2018 – Aug. 2018*

- **Backend Software Development:** Migrated core functionality of the Company Financials Core Algorithms Library from C++ to Spark in Scala.

USC Spatial Sciences Institute Los Angeles, CA
Research Assistant · Advisor: Yao-Yi Chiang *Aug. 2016 – Aug. 2018*

- **Timeseries Data Analysis:** Analyzed dynamic and large-scale transportation sensor data of buses in LA.

Professional Services

- **Conference Reviewer:** IROS 2022, CoRL 2023, ICRA 2024
- **Journal Reviewer:** RA-L

Teaching

- Introduction to Machine Learning (10-301/10-601), Teaching Assistant, CMU *Fall 2021*
- Mathematical Statistics (36-700), Teaching Assistant, CMU *Spring 2021*
- Introduction to Computer Systems (CSCI 356), Teaching Assistant, USC *Spring 2020*
- Discrete Methods in Computer Science (CSCI 170), Teaching Assistant, USC *Spring, Fall 2019*
- Introduction to Algorithms and Theory of Computing (CSCI 270), Teaching Assistant, USC *Fall 2018*